

Web Scraper System — Full Technical Documentation

1. System Overview

This document explains the structure and logic of the Playwright-based web scraper.

It extracts company contact data (email, phone, social links) and updates Google Sheets automatically.

2. High-Level Architecture

Main components:

1. Google Sheets integration
2. Regex-based data extractors
3. Playwright browser engine
4. Resume controller and error handling

3. Google Sheets Connection

The scraper uses a Google Service Account JSON key.

The key authenticates the script without requiring user login.

The service account must be added as an editor to the target Google Sheet.

4. Start-Row Selection Logic

Before scraping, the script asks:

- Continue from last unfinished row
- Start from row 5

Rows are considered unfinished if Column M contains: empty, Not Found, Error.

5. Email Extraction Logic

The email extractor removes:

- Tracking/logging domains (sentry.io, wixpress.com)
- File-extension-type junk (.png, .jpg, .css)
- Long hex IDs
- Invalid unicode fragments (u003e)
- Numeric-only domains

It keeps only realistic business emails.

6. Phone Extraction Logic

Phone filtering rules include:

- Must begin with +, 0, or toll-free 8
- Must be 8–15 digits
- Reject plain 8-digit strings
- Reject VAT/company number patterns
- Reject year-like strings (2024, 2025)
- Reject malformed brackets
- At least one digit group must have 4+ digits

7. Social Extractors

Extracts:

- LINE
- Facebook
- Instagram
- Telegram
- Contact Form URLs

8. Playwright Browser Workflow

1. Launch Chromium with desktop user-agent
2. Retry failed loads
3. Extract visible text + HTML
4. Process via extractors
5. Write results to Google Sheets

9. Sheets Update Logic

Columns M–S store:

- Email
- Phone
- LINE
- Facebook
- Instagram
- Telegram
- Contact Form

10. Error Handling

After two load failures, the row is marked as an error but scraping continues.
If visible text extraction fails, the scraper falls back to raw HTML.

11. Limitations

- Regex-based detection has natural limits
- Some sites block bots
- Fully dynamic sites may need additional waits

12. Future Improvements

- CAPTCHA bypass
- ML-based phone validation
- Distributed scraping
- Crawl sitemap automation